POKHARA UNIVERSITY

Level: Master | Semester: Spring | Year : 2025
Programme: ME/MSc Computer | Full Marks : 100
Course: Data Structure and Algorithm Design (New) | Pass Marks : 60
Time : 4 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What is the principle behind the greedy algorithm? Explain with the example of fractional knapsack problem in detail. 7

   b) What is the idea behind the backtracking algorithm? Explain with the example of 8-queen problem in detail. 8

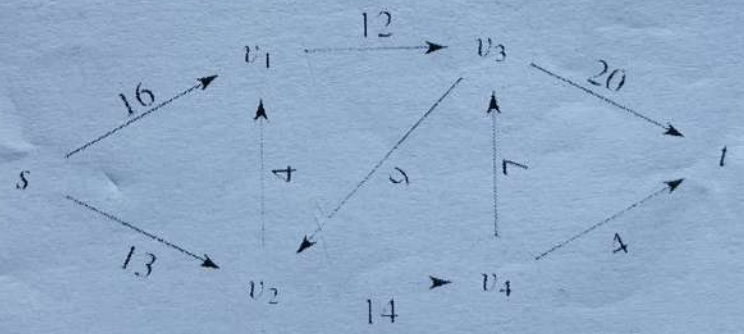
OR

What is asymptotic analysis of algorithms? Why is it better than the empirical approach for determining efficiency of algorithms? Discuss in detail.

2. a) What are the limitations of B Tree? How does the B+ Tree resolve these problems? Explain in detail with appropriate examples. 8

   b) What is relaxation in solving shortest path problem? Explain its role in Bellman-Ford algorithm with example in detail. 7

3. a) Apply the Ford-Fulkerson algorithm to find the maximum flow in the given network: 7



   b) What are KD trees? Why are they beneficial? Explain in detail with appropriate example. 8

4. a) Define locality of reference. Why is the locality of reference important for CPU performance? Explain in detail with example. 8

   b) What are cache-efficient algorithms? Explain the cache-conscious merge sort or cache-conscious matrix multiplication in detail. 7

5. a) What are profiling tools? Explain any one profiling tool for analyzing memory and cache efficiency. 8

b) What is consistent hashing? How does it resolve the problem during searching when the node or cache are added or removed in the distributed system? Explain in detail. 7

6. a) How does bucket sort work? Explain in detail with appropriate example. Also analyze its cost. 8

OR

What is task parallelism? Use any tool of task parallelism like OpenCilk or OpenMP to implement the parallel merge sort.

b) What are the limitations of Embedded system? Explain how the static Huffman coding improve the performance in the embedded system. 7

7. Write short notes on: (**Any two**) 2×5

a) Distributed Hash Table

b) 2-way external merge sort

c) Memoization in dynamic programming

POKHARA UNIVERSITY

Level: Master | Semester: Spring | Year : 2025
Programme: ME/MSc Computer | Full Marks : 100
Course: Advanced Computer Architecture (New) | Pass Marks : 60
Time : 4 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Why is parallel programming essential in modern computing? Discuss the key characteristics of parallel algorithms and explain how they influence the design and efficiency of parallel systems. 8

   b) Differentiate between data and instruction. Briefly explain about the architectural classification of parallel processing. 7

2. a) Explain the terms: 8
   i. Memory Capacity Planning
   ii. Virtual Memory Technology

   b) How does the Least Recently Used (LRU) page replacement algorithm differ from FIFO and Optimal? Give an example. 7

3. a) Discuss the role of Cannon's Algorithm and systolic arrays in optimizing matrix multiplication for high-performance computing applications. 7

   b) Explain pipeline hazards and how they affect instruction pipeline performance. 8

OR

Dynamic instruction scheduling reduces pipeline hazards and improves microprocessor performance? Justify.

4. a) How does multithreading improve CPU utilization compared to multiprocessing? Provide real-world scenarios. 7

OR

Describe principles of thread-level parallelization. Discuss mutexes, semaphores, and barriers with relation to grid computing.

   b) Explain the terms: 8
   i. Cluster Computing
   ii. Neural Computing

iii. Grid Computing

5. a) Why is cache coherence important in multi-core systems, and what protocols are commonly used? 7

b) What is GPGPU computing, and how is it different from traditional GPU usage? 8

6. a) Discuss the classification of multiprocessor operating systems. How do symmetric and asymmetric multiprocessor OS differ in design and functionality? 8

b) What is PThread, and why is it useful in shared memory systems? Explain. 7

7. Write short notes on: (**Any two**) 2×5

a) Feng's Classification

b) Vectorization

c) Latency hiding techniques

POKHARA UNIVERSITY

Level: Master | Semester: Spring | Year : 2025
Programme: ME/MSc Computer | Full Marks : 100
Course: Operating System Design and Implementation (New) | Pass Marks : 60
Time : 4 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) How do System call and Library call interact in an Operating System? Discuss the exokernel architecture of operating system and how it differs from traditional operating system kernels. 8

   b) How is multiprocessor different from multicore system. Explain the concept of multi-core process scheduling with and the role of CPU affinity in optimizing performance with necessary illustrations. 7

2. a) Explain the necessity of thread synchronization and describe how semaphores, along with their structure and operations, can be used to achieve synchronization. Illustrate your answer with a suitable practical example. 8

   b) Discuss the significance of wait-for graphs in the detection of deadlocks. List out the various categories of distributed deadlock detection algorithms and describe one of them in detail. 7

3. a) What are the key benefits of virtualization in computing? Compare and contrast the two main types of hypervisor architectures. 7

   b) Define Virtual memory. What are the performance implications of choosing a small page size versus a large page size in virtual memory systems? Discuss in terms of fragmentation, page table size, and TLB performance. 8

4. a) Discuss the limitations of traditional file systems in handling massive data volumes. Explain the architecture of GFS in detail. 8

   b) Differentiate between SSD and HDD. What are the key components involved in an SSD, and how do they work together to deliver high-speed data storage and retrieval? 7

5. a) What is SELinux, and how does it improve security in Linux based operating system? What are the four components of SELinux and write role of each component. 8

b) What is embedded System? List the features of Embedded Operating System. Discuss the key design principles of energy-efficient operating systems. 7

6. a) For three tasks: T1(C1=2, P1=5), T2(C2=3, P2=10), T3(C3=4, P3=20), calculate the hyperperiod, construct the RMS schedule for one hyperperiod, and determine the completion time of each task instance. 7

b) Explain the role of Kubernetes in Cloud Operating System. Discuss the concept of fault tolerance and load balancing in distributed system. 8

7. Write short notes on: (**Any two**) 2×5

a) Cache-aware Memory Management

b) Transparency in Distributed System

c) Mitigation Techniques for Side-channel Attacks

POKHARA UNIVERSITY

Level: Master
Semester: Spring
Year : 2025
Full Marks : 100
Programme: ME/MSc Computer
Pass Marks : 60
Course: Research Techniques (New)
Time : 4 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define research. How can we relate a research work with everyday thinking? Explain with a suitable example. 7

   b) Explain the different types of research with appropriate examples in brief. 8

2. a) What is a literature review? Explain the purpose of literature review in detail. 7

   b) What are the possible resources for the literature review? Explain in detail. 8

3. a) Explain the role of methodology in a research work. 7

   OR

   What are the components of a research methodology? Explain in brief.

   b) What is research design? Explain the experimental research design in detail. 8

4. a) What are the different data collection methods? Explain in brief. 8

   b) What is quantitative data analysis? Explain the quantitative data presentation methods in detail. 7

5. a) Explain any one analysis technique for quantitative data. 8

   OR

   Explain qualitative data analysis in detail.

   b) What is textual data analysis in qualitative data analysis? Explain in detail. 7

6. a) What are the rights of the people that are directly involved in the research? Explain the responsibilities of an ethical researcher. 7

   b) What are the different written presentations of a research work? Explain how the research work is presented in academic papers. 8

7. Write short notes on: (**Any two**) 2×5

a) Referencing and Citation

b) Copyright and Patent

c) Posters Presentation

POKHARA UNIVERSITY

Level: Master | Semester: Spring | Year : 2025
Programme: ME/MSc Computer | Full Marks : 100
Course: Software Engineering (New) | Pass Marks : 60
Time : 4 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) How do Waterfall, Iterative, Agile, and DevOps/CI-CD models differ in handling changing requirements, risk, resource use, and scalability? What are the trade-offs between flexibility, predictability, and speed, and how can organizations choose the right model based on project complexity, uncertainty, and stakeholder needs? 10

   b) What is Software Engineering? Can it be considered a true engineering discipline? Justify. 5

2. a) How do project organization factors such as roles, task distribution, and scheduling interact with communication practices including meetings, tools, and informal exchanges to shape the success or failure of software projects, and what strategies can ensure their effective alignment? 8

   b) What is Docker, and in what ways does it simplify the deployment and management of applications across diverse environments? 7

3. A software development team is tasked with building an online bookstore platform that allows customers to browse and purchase books, track their orders, and make secure payments online. The platform should also support bookstore owners in managing book inventory, prices, and promotions, while providing delivery personnel with tools to process and deliver orders to customers. 15

   a) Who are the key actors and stakeholders involved in this system? Provide a list and briefly describe their roles and interests.

   b) Create two scenarios: one for a customer purchasing a book and another for a bookstore owner adding a new book to the catalog. Based on these scenarios, draft initial use cases for each scenario.

   c) Draw a use case diagram that shows the relationships between the actors (customer, bookstore owner, delivery personnel) and the use

cases (purchase book, manage catalog, track delivery). Explain how these relationships help the system satisfy the needs of all stakeholders.

4. a) Using the Library Management System as a case, analyze how class diagrams and sequence diagrams can be applied to model the system's structure and behavior. How do these diagrams help in understanding the relationships between classes and the flow of interactions during key processes such as borrowing a book? What challenges might arise when creating these diagrams, and how can they be addressed to ensure accurate system design? 10

OR

Using the Library Management System as a case, explain how use case diagrams and activity diagrams can be used to capture the system's requirements and workflows. How do these diagrams assist in understanding user interactions and the step-by-step flow of processes such as returning a book? What difficulties might be faced when designing these diagrams, and how can they be overcome to ensure clarity and completeness?

b) Explain the role of abstraction, inheritance, and polymorphism in object-oriented paradigm? Provide a simple example for each concept to explain how they are used in software development. 5

5. a) Design patterns are widely used to promote software reuse and improve maintainability in software systems. In this context, answer the following: 8

   i. How does delegation reduce tight coupling between classes? Give an example of a design pattern that applies delegation.

   ii. What problem can occur if implementation inheritance is overused, and how does following the Liskov Substitution Principle (LSP) help overcome it?

b) How does effective stakeholder communication influence the success of the requirements gathering phase in software projects? Discuss the risks of inadequate stakeholder participation, and suggest strategies that can be used to improve collaboration and ensure the completeness of requirements. 7

6. a) What is unit testing, integration testing, and system testing? Describe its purpose, the stage at which it is performed? Also, 7

discuss how this contribute to ensure the quality and reliability of a software system.

b) Explain the importance of configuration items, baselines, and change management processes such as handling change requests and version/release management. How do these practices ensure controlled and systematic updates to the software system? Provide examples to support your answer. 8

OR

Software Quality Assurance (SQA) plays a vital role in ensuring that software systems meet defined quality standards. Explain the objectives of SQA and discuss the key activities involved, such as reviews, audits, testing, and process monitoring. How do these activities contribute to reducing defects, improving reliability, and ensuring customer satisfaction in software projects?

7. Write short notes on: **(Any two)** 2×5

a) SOA Architecture

b) IOT based Software Engineering

c) Secure Coding Practices